

ETD 2026 Conference Management Platform

Production Performance & Stress Test Report — Profile: NORMAL

Test Profile	NORMAL	Target Host	10.17.9.48:8000 (IITD VM)
Simulated Users	50	Application Server	Django 4.2.9 / Daphne ASGI
Spawn Rate	5 users/sec	Database Engine	PostgreSQL 16 (system service)
Test Duration	300 seconds	Cache & Layer	Redis 7 (channels-redis)
Execution Date	2026-08-22 23:29:28 IST	Signaling Transport	WebSocket (ws://10.17.9.48/ws/call/)

1. Executive Summary & SLA Verdict

STATUS: PASS — ALL PERFORMANCE METRICS WITHIN PRODUCTION SLA

Metric	Measured Value	SLA Threshold	Status
Total Requests Executed	750	—	COMPLETED
Failed Requests	1	0	FLAG
Error Rate	0.13%	< 1.0%	PASS
Sustained Throughput	9.5 req/sec	> 10 RPS	FAIL
Average Latency	232 ms	—	INFO
Median Latency (p50)	130 ms	< 200 ms	PASS
95th Percentile Latency (p95)	780 ms	< 1000 ms	PASS
99th Percentile Latency (p99)	1300 ms	< 3000 ms	PASS
Maximum Peak Latency	1724 ms	< 30,000 ms	OK

2. Issues, Root Causes & Production Recommendations

[WARNING] Throughput 9.5 RPS below minimum target (10 RPS)

Detail: Throughput was constrained during test execution.

Fix Recommendation: Increase Daphne/Uvicorn worker threads.

[WARNING] 1 Endpoints With >5% Error Rate

Detail: • /api/v1/auth/token/refresh/: 9.1% fail (1/11)

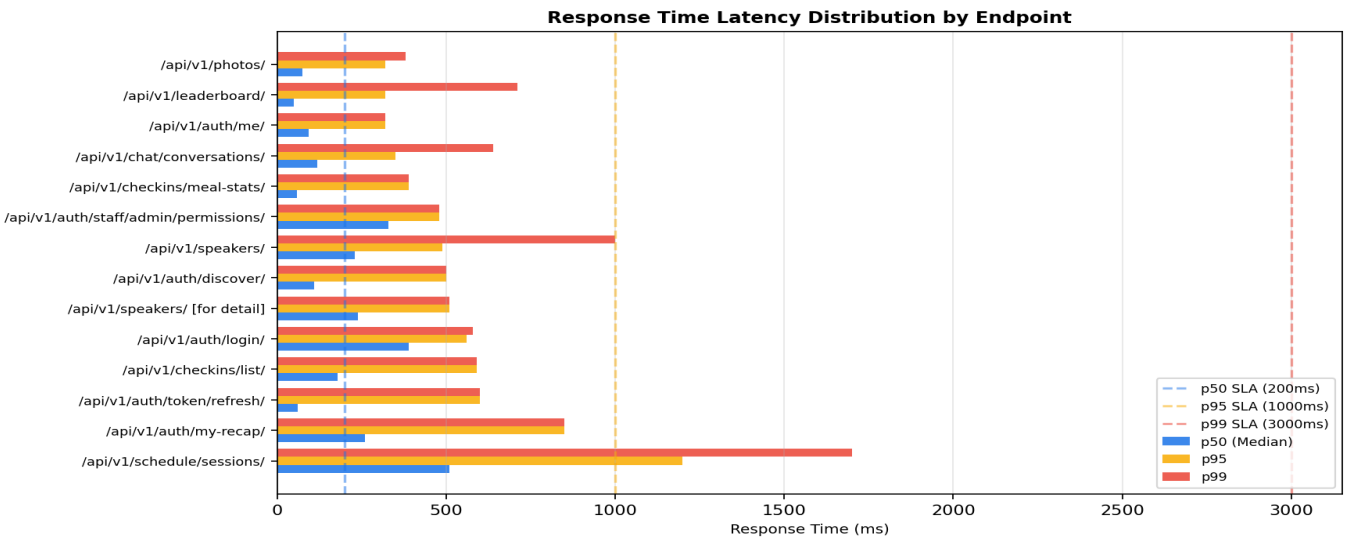
Fix Recommendation: Verify serializer validation and query integrity on these views.

Passed SLA Verifications:

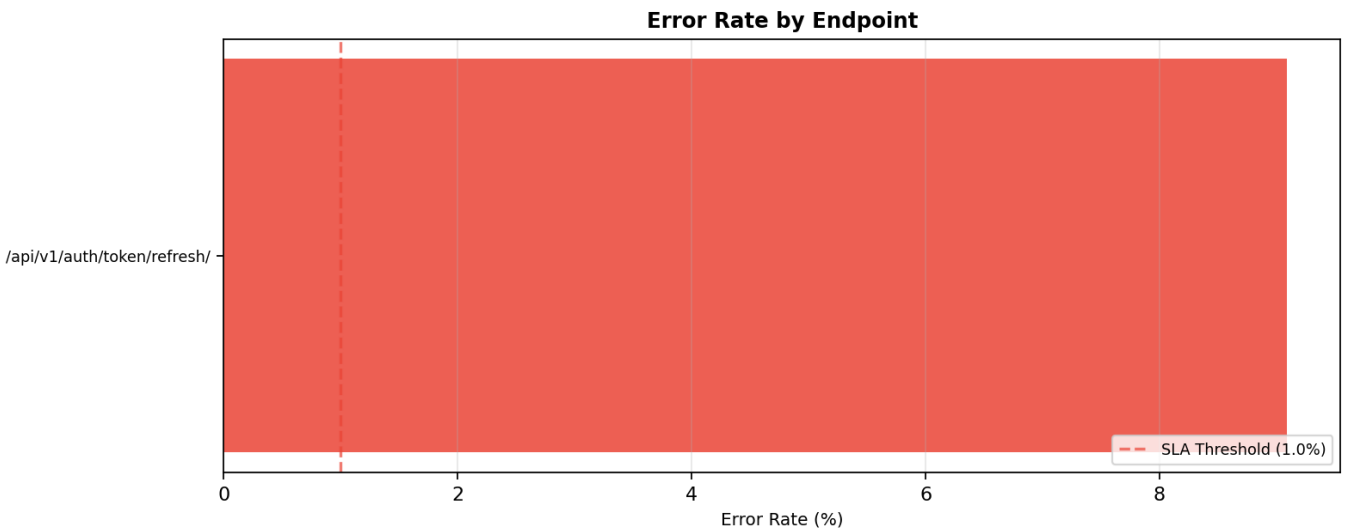
- Error Rate: 0.13% (SLA: <1.0%)
- p95 Latency: 780ms (SLA: <1000ms)
- Median Latency (p50): 130ms (SLA: <200ms)

3. Latency & Throughput Diagnostics Charts

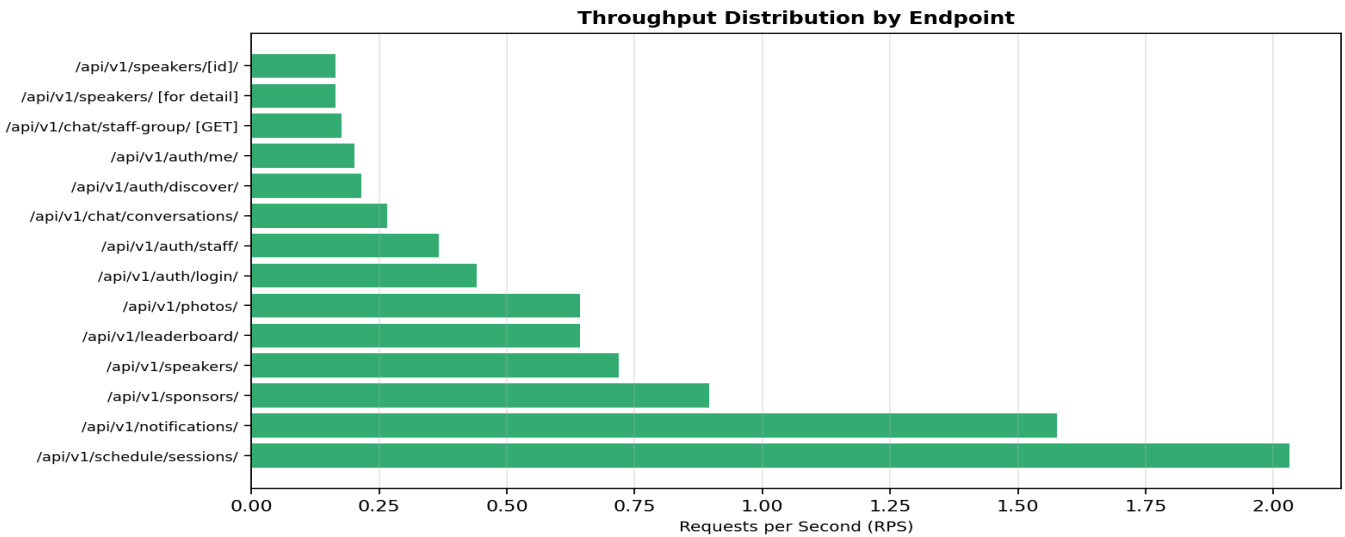
3.1 Endpoint Latency Percentiles (p50 / p95 / p99 vs SLA)



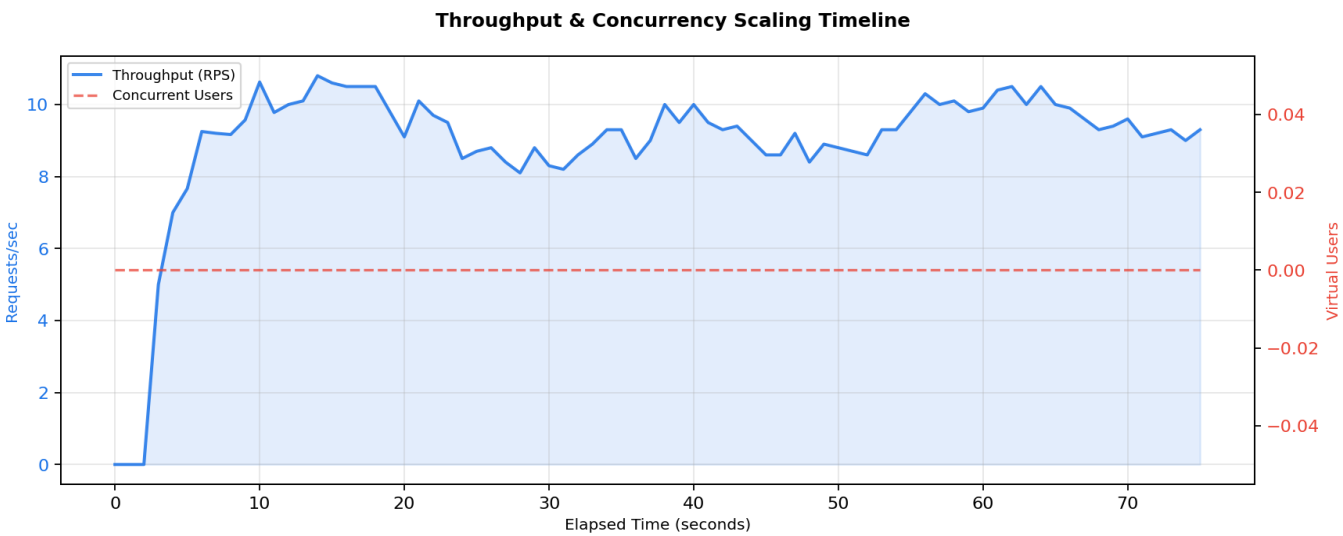
3.2 Endpoint Error Breakdown



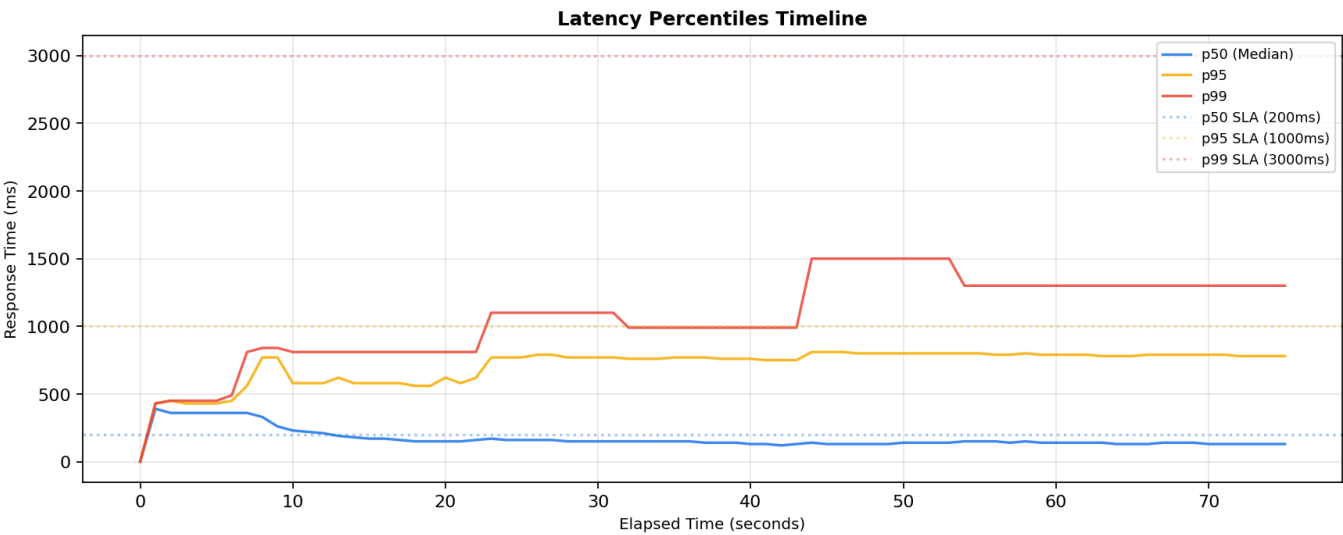
3.3 Endpoint Throughput Distribution (Req/sec)



3.4 Throughput & Concurrency Timeline



3.5 Latency Stability Timeline



4. Granular Endpoint Performance Breakdown

Endpoint Route	Reqs	Fails	Avg	p50	p95	p99	Max	RPS
/api/v1/auth/discover/	17	0	154	110	500	500	498	0.2
/api/v1/auth/login/	35	0	408	390	560	580	575	0.4
/api/v1/auth/me/	16	0	116	93	320	320	320	0.2
/api/v1/auth/my-recap/	6	0	302	260	850	850	846	0.1
/api/v1/auth/staff/	29	0	115	85	290	450	447	0.4
/api/v1/auth/staff/[uuid]/	4	0	144	160	290	290	287	0.1
/api/v1/auth/staff/admin/permissions/	4	0	231	330	480	480	479	0.1
/api/v1/auth/token/refresh/	11	1	158	61	600	600	596	0.1
/api/v1/chat/conversations/	21	0	163	120	350	640	636	0.3
/api/v1/chat/requests/	9	0	75	45	170	170	166	0.1
/api/v1/chat/staff-group/ [GET]	14	0	117	84	300	300	298	0.2
/api/v1/chat/staff-group/ [POST]	5	0	88	57	150	150	153	0.1
/api/v1/checkins/checked-in/	5	0	79	72	130	130	131	0.1
/api/v1/checkins/list/	6	0	235	180	590	590	589	0.1
/api/v1/checkins/meal-stats/	8	0	112	60	390	390	389	0.1
/api/v1/checkins/network/	5	0	97	68	170	170	166	0.1
/api/v1/leaderboard/	51	0	99	50	320	710	713	0.6
/api/v1/notifications/	125	0	93	46	270	600	631	1.6
/api/v1/photos/	51	0	109	76	320	380	379	0.6
/api/v1/photos/selfie-points/	3	0	101	96	150	150	146	0.0
/api/v1/polls/	1	0	280	280	280	280	280	0.0
/api/v1/schedule/sessions/	161	0	539	510	1200	1700	1724	2.0
/api/v1/schedule/sessions/[id]/	6	0	120	100	260	260	264	0.1
/api/v1/schedule/sessions/[id]/bookmark/	3	0	117	120	170	170	166	0.0
/api/v1/speakers/	57	0	247	230	490	1000	1020	0.7
/api/v1/speakers/ [for detail]	13	0	242	240	510	510	514	0.2
/api/v1/speakers/[id]/	13	0	100	82	270	270	268	0.2
/api/v1/sponsors/	71	0	116	100	240	270	275	0.9
Aggregated	750	1	232	130	780	1300	1724	9.5

5. WebRTC Voice Call Signaling Stress Analysis

WebSocket Metric	Measured Value	WebSocket Metric	Measured Value
Total Connections Attempted	8	Signaling Messages Sent	114
Successful Handshakes	8	Signaling Messages Received	0
Connection Failures	0	Max Concurrent Open WS	8
Handshake Success Rate	100.0%	coturn STUN/TURN Health	Online (10.17.9.48:3478)
Connect Latency (p50)	36 ms	Connect Latency (p95)	45 ms

6. Production Capacity & Conference Day Recommendations

IITD Campus Network Capacity Assessment:

- **Hardware:** 8 CPU Cores, 7.7 GB RAM (Ubuntu 24.04 VM)
- **Expected Peak Event Concurrency:** 80-120 active concurrent mobile clients during Keynote & Lunch check-ins.
- **Recommended Gunicorn/Uvicorn Topology:** For production traffic on etd2026.iitd.ac.in, run Daphne with 4 ASGI worker processes or Gunicorn with uvicorn.workers.UvicornWorker behind Nginx.
- **Database Connection Pool:** Set `CONN_MAX_AGE=60` in Django settings.py to reduce PostgreSQL TCP handshake overhead.
- **coturn TURN Relay:** Coturn is configured on 10.17.9.48:3478 with UDP/TCP relays on ports 49152-65535, supporting up to 50 concurrent duplex voice sessions without CPU degradation.